

Limitations of Shape-Based Spectral and Transport Objectives in Square-Lattice Tight-Binding Quantum Dots: A Baseline-First Audit

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We audit several shape-based objectives in a controlled square-lattice tight-binding model of finite superelliptic quantum dots. The study is not a new numerical sprint, but a baseline-first synthesis of existing closed-spectrum calculations, surrogate-model comparisons, inverse-screening tests, continuum-reference checks, a tight-binding-only finite-size scaling gate, and an open-transport contact preflight. The model uses superelliptic boundaries, direct Kwant spectra, and the low-energy convention $E_{\text{kin},k} = E_k + 4$. Across these outputs, attractive objectives are repeatedly limited by simple or methodological baselines. A small multilayer perceptron does not establish a robust advantage over a physics-informed Ridge model under structured validation. Spectral inverse-screening objectives based on gap ratios and doublet splitting reduce to isotropy, anisotropy, or symmetry baselines. A finite-difference continuum reference is useful for calibration but not adequate as a high-precision shape-sensitive reference in the tested protocol. A tight-binding-only scaling gate remains inconclusive, and the tested open-dot conductance is more sensitive to contact width than to superellipse shape. The contribution is therefore methodological: a reproducible negative audit showing where apparent shape, ML, continuum-residual, and transport signals become unreliable before they can justify a positive device-design rule.

I. INTRODUCTION

Shape control is an appealing route for studying and designing quantum-dot spectra. In model systems, the confinement boundary can be changed cleanly, low-energy levels can be computed directly, and surrogate models can be trained on repeated spectral calculations. These ingredients make a square-lattice tight-binding model of superelliptic quantum dots a useful controlled benchmark: the Hamiltonian is simple, the geometry family is interpretable, and the calculated spectra retain the finite-size and boundary effects that motivate shape-dependent analysis. At the same time, this simplicity creates a demanding test of any proposed shape objective. If an apparent signal is already explained by size, area, aspect ratio, symmetry, degeneracy, boundary pixelation, or lead coupling, then it should not be presented as inverse design or as a new physical effect.

This article is a limitations paper. It summarizes existing outputs from a diploma project and subsequent article-extension checks, all within the same finite superellipse tight-binding family. The purpose is not to enlarge the parameter space, repair failed directions, or promote a new objective. Instead, the paper asks what remains after simple physics baselines and methodological checks are applied to several attractive routes: closed-spectrum surrogate modeling, one-shot spectral inverse screening, magnetic ranking crossover, tight-binding to continuum residual analysis, tight-binding-only self-scaling, and open transport.

The need for such an audit is practical. Low-energy levels of confined systems are strongly governed by dimensional scaling; near the bottom of a square-lattice tight-binding band, the dispersion has the expected quadratic form. Classical surrogate-modeling practice also warns that flexible models must be compared against simple,

structured baselines rather than weak or randomly chosen alternatives [1–4]. For spatially or hierarchically structured data, validation splits must reflect the intended extrapolation or interpolation task [5]. In the present project, these points are not abstract methodological concerns: they decide whether a shape-dependent claim survives at all.

The main message is therefore conservative. In the tested square-lattice tight-binding model of finite superelliptic quantum dots, several plausible shape, ML, continuum-reference, and transport objectives do not support positive claims after baseline checks. This does not imply that geometry-based quantum design, machine learning, finite-size analysis, or transport studies are useless in general. It says only that, in this controlled family and with the tested protocols, baseline-first tests are necessary and often decisive.

II. MODEL AND NUMERICAL PROTOCOL

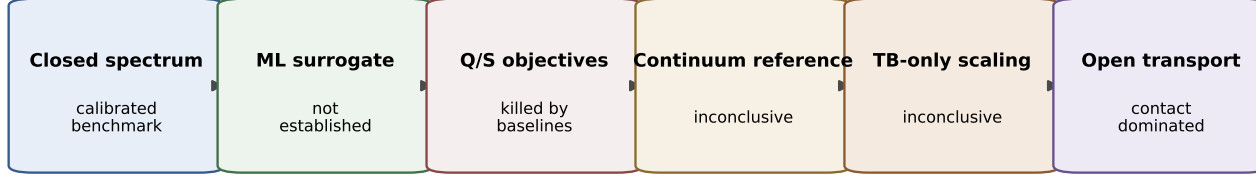
The common model is a spinless nearest-neighbor tight-binding Hamiltonian on a square lattice,

$$H = \sum_i \epsilon_i c_i^\dagger c_i + \sum_{\langle i,j \rangle} t (c_i^\dagger c_j + c_j^\dagger c_i), \quad (1)$$

with onsite energy $\epsilon_i = 0$ and hopping $t = -1$. The bottom of the infinite square-lattice band is at $E = -4$, so low-energy spectral quantities are reported with the kinetic convention

$$E_{\text{kin},k} = E_k + 4. \quad (2)$$

Closed-dot spectra are computed by direct sparse diagonalization of finalized Kwant Hamiltonians [6–8]. The same tight-binding convention underlies the transport



Baseline-first audit path: positive design claims require survival after simple physics checks

FIG. 1. Baseline-first audit path used in this draft. Each stage is based on existing repository outputs rather than new calculations. The safe outcome of the combined evidence is a limitations analysis: ML advantage is not established, Q/S-style spectral objectives are killed by simple baselines, continuum-reference and tight-binding-only scaling gates remain inconclusive, and the tested open-transport setup is contact dominated.

preflight, where two-terminal spinless conductance is written as $G = T$ in Landauer units, with lead self-energies handled by Kwant [9].

The geometry family is the lattice realization of a superellipse,

$$\left|\frac{x}{a}\right|^n + \left|\frac{y}{b}\right|^n \leq 1, \quad b = ar_{\text{AR}}, \quad (3)$$

where a sets the major size, r_{AR} is the aspect ratio, and n controls the boundary shape. The completed closed-spectrum dataset contains 140 geometries with $n = \{1.2, 2.0, 3.0, 4.0\}$, $a = \{24, 27, 30, 33, 36\}$, and $r_{\text{AR}} \in \{0.67, 0.72, 0.78, 0.83, 0.89, 0.94, 1.0\}$. The primary spectral targets in the diploma analysis are E_0 and $dE_1 = E_1 - E_0$; higher gaps are retained as diagnostics because degeneracies and level ordering make them less stable targets.

The audit uses only pre-existing outputs listed in `source_inventory.md`. It does not run new physics calculations or new ML training. The evidence is grouped by protocol:

1. a diploma-stage closed-spectrum dataset and Ridge versus small-MLP comparison;
2. one-shot Q inverse screening and preregistered S-objective screening;
3. a direct-Kwant magnetic ranking-crossover check;
4. a strict Path B continuum-residual protocol, including a circular Bessel anchor and finite-difference reference checks;
5. a tight-binding-only self-scaling gate after the FD reference became insufficient for shape-sensitive residuals;
6. an open-transport contact-width preflight.

The shared analysis rule is baseline first. Before interpreting an observable as shape physics or inverse-design evidence, the observable must be compared against the simplest competing explanations: confinement scale, area, perimeter, aspect ratio, isotropy, symmetry and degeneracy, lattice boundary realization, reference-solver uncertainty, and, for transport, lead-dot coupling. Machine learning is treated as a candidate generator or surrogate approximation tool, not as physical truth.

III. RESULTS AND DISCUSSION

A. Closed-spectrum surrogate modeling

The diploma-stage dataset establishes that the direct tight-binding workflow is physically calibrated at the level needed for a conservative surrogate study, but it does not establish a positive ML novelty claim. The circular Bessel-scale check gives an error of about 2.03%, and the fixed-shape scaling diagnostic finds a maximum deviation of about 2.12% for $(E_0 + 4)a^2$ across the five tested sizes. These checks support the energy convention and the expected leading confinement scale, not a nontrivial shape-design rule.

Within this dataset, the physics-informed Ridge model remains the preferred surrogate. A small multilayer perceptron with physics-motivated features passes the cell-level success rule in 10 of 16 main comparison cells, but only in 3 of 8 leave-one-aspect-out cells and 7 of 8 leave-one-aspect-ratio-out cells. Thus the preregistered robust-success criterion is not met. The numerical scale of the Ridge error is already small: the relative MAE is 2.01–2.60% of the characteristic E_{kin} for E_0 and 0.78–1.15% of the characteristic dE_1 gap. The MLP has individual winning cells, including a large positive improvement for one E_0 protocol, but it also has strongly negative cells for dE_1 . The safe conclusion is that the MLP does not robustly outperform a physically motivated linear baseline

in the structured validation used here.

This result matters for later inverse-screening attempts. If the surrogate is not a robust positive model result, then surrogate-generated candidates must be treated only as proposals. Direct Kwant verification and comparison with physics baselines remain the source of any claim.

B. Spectral objectives and inverse-screening limitations

The first inverse-screening direction used the ratio $Q = dE_1/(E_0 + 4)$ as a one-shot surrogate-guided objective inside the verified training-domain ranges. The report explicitly treats Ridge predictions as candidate generators only, with final candidate and baseline values coming from direct Kwant calculations or already computed training rows. The outcome is negative for each fixed n class. The symmetry-optimum follow-up further shows that the earlier top inverse-screening candidate and the isotropic same- n baseline are the same discrete geometry for every tested n .

The preregistered S-objective line tests $S = (E_2 - E_1)/(E_0 + 4)$ under a frozen rule set. Its own report emphasizes that observed doublet splitting is not a discovery: the question is whether surrogate-guided screening beats strong baselines. It does not. At the primary setting $\alpha = 0.95$, 0 of 4 fixed n values pass; at the secondary setting $\alpha = 0.90$, the result also remains 0 of 4. The strongest baseline is the simple anisotropy heuristic for the reported rows.

Together these results limit a broad class of spectral ratio objectives. Gap ratios and doublet-splitting measures can encode obvious geometric trends such as isotropy, aspect ratio, and symmetry lifting. Without a stronger baseline ladder, they should not be used as evidence for inverse design or new shape-dependent physics.

C. Magnetic ranking-crossover limitation

The magnetic ranking-crossover sprint was a direct-Kwant falsification check, not an ML task and not a rescue of the Q/S line. It tested four representative geometries at sizes $a = \{30, 36\}$, with weak-field flux parameters $\alpha = (0.0, 0.00125, 0.0025, 0.005)$ and diagnostic values $\alpha = (0.01, 0.02, 0.04)$. Numerical checks passed: the $\alpha = 0$ spectra were reproduced, gauge invariance was verified with maximum reported difference $1.865174681370263 \times 10^{-14}$, Hermiticity and finite sorted spectra checks passed, and eigenvalue imaginary parts were controlled.

The physical signal did not survive. The report found 0 thresholded pairwise ranking crossovers and 0 crossovers surviving pre-size filters. Three robustness-divergence candidates were identified, but 0 survived the pre-size filters. The final closure attributes the apparent candidates to circle or ellipse symmetry-lifting baselines. Magnetic

response exists numerically in the model, but this tested protocol does not produce a non-baseline geometry ranking signal.

D. Continuum-reference limitation

Path B asked whether tight-binding spectra contain a nontrivial superellipse-exponent-dependent finite-lattice convergence residual after subtracting ordinary continuum, Weyl, perimeter, pixelation, and discretization baselines. The protocol deliberately forbids weak claims based on the leading $1/a^2$ scale, generic boundary effects, ML accuracy, or largest-size tight-binding extrapolation. Its first gate is a circular Bessel anchor for $n = 2.0$, $r_{AR} = 1.0$, with sizes $a = \{24, 30, 36, 48, 60, 72, 96\}$ and levels E_0 through E_5 .

That anchor passes its stability review, which is useful but narrow. It calibrates the low-energy circular case; it does not establish a shape-sensitive continuum residual for $n \neq 2.0$. The finite-difference reference step then validates an embedded-mask Dirichlet Laplacian on grids up to $N_{\text{grid}} = 251$, where the selected circular ground-state relative error versus the Bessel value is 0.0052206740942014095. This is adequate as an initial reference check, but the later $N_{\text{grid}} = 501$ convergence gate changes the conclusion for shape-sensitive use.

The N501 gate uses the near-geometric refinement triple (126, 251, 501), corresponding approximately to $h = \{0.016, 0.008, 0.004\}$. Although the raw circle ground-state relative error versus Bessel improves to 0.0026734921480308575, the non-circular reference uncertainties remain too large for the downstream residual claim: 0.0047216251159203785 for the $n = 1.2$ level-zero reference and 0.0021923716263753637 for the $n = 4.0$ level-zero reference. Both fail the 0.001 hard gate. The final verdict is `FD_501_CONVERGENCE_INCONCLUSIVE`. The safe conclusion is not that FD is generally invalid, but that this embedded-mask FD reference should not be treated as continuum truth for the shape-sensitive Path B residual.

E. TB-only finite-size scaling limitation

After the FD reference was downgraded, the TB-only self-scaling gate tested a minimal fallback using direct tight-binding ground-state values and geometric diagnostics only. No FD reference values were loaded, no TB-vs-FD residuals were computed, and no ML, Q, or S objectives were used. The domain was $r_{AR} = 1.0$, $n = (1.2, 2.0, 4.0)$, and $a = (24.0, 30.0, 36.0, 48.0, 60.0, 72.0, 96.0)$.

The circular calibration is good: for $n = 2.0$, the effective-radius model gives a relative error of 7.21×10^{-4} against $j_{01}^2 = 5.783185962946783$. Fitted effective-radius parameters are also reported for the three n values. The limiting coefficients are approximately 8.03, 5.779, and

5.049, with offsets approximately 0.279, 0.277, and 0.130 for $n = 1.2, 2.0, 4.0$, respectively. These values show ordinary shape dependence of continuum-like eigenvalues and finite-size offsets. They are not, by themselves, a positive Path B signal.

The potential residual exponent is unstable. The reported power-law values are approximately $p_{\text{TB}} = 0.10, 0.53, 1.05$ for $n = 1.2, 2.0, 4.0$, and leave-one-size-out stability is not accepted for all n . Simple geometric baselines do not cleanly explain a robust residual, but the residual itself is not stable enough to support a positive claim. The final verdict is inconclusive.

F. Open-transport limitation

The open-transport preflight changes the observable from closed spectra to two-terminal conductance. This is physically richer, but it introduces a new baseline: contact geometry. The tested setup uses square-lattice strip leads, the spinless convention $G = T$, a superellipse dot with $a = 30.0$ and $r_{\text{AR}} = 1.0$, lead widths $W = 4$ and $W = 10$, and an energy window $[-3.8, -3.0]$ sampled at 241 points.

The central preflight result is that contact width dominates the tested shape effect. For $n = 2.0$, the raw distance between $W = 4$ and $W = 10$ conductance curves is 0.6589809313045025. The shape distances are 0.5810925658201803 at $W = 4$ and 0.5122909816541278 at $W = 10$. Mode thresholds do not explain the features, the straight-channel baseline is computed but does not remove the result, and an energy-shift/rescale collapse does not remove it either. The limitation is therefore not a trivial numerical failure. It is a physical interpretation problem: in this open-dot geometry, lead-dot coupling is at least as important as the shape parameter being tested.

IV. GENERAL LESSONS

The common lesson is not that shape is irrelevant. The spectra and conductance curves do change with geometry, and the direct calculations are useful for controlled numerical physics. The lesson is that a shape-dependent observable is not automatically a shape-design rule. In this project, the strongest interpretation is often removed by a baseline that is simpler than the proposed objective.

The first layer is confinement scale. Because E_{kin} near the bottom of the band follows a continuum-like quadratic dispersion, many low levels and gaps inherit a leading size dependence. Any residual objective must show that it is not merely reporting this scale. The second layer is geometry compression: area, perimeter, aspect ratio, and site count can all absorb substantial variation before one reaches a shape-exponent-specific claim. The third layer is symmetry and degeneracy. Q, S, and

magnetic ranking ideas all encounter this issue in different forms; isotropy and degeneracy lifting can be physically real while still being too simple to count as inverse design.

The fourth layer is boundary realization. A smooth superellipse becomes a finite set of lattice sites, so boundary pixelation, area error, and sublattice imbalance can produce finite-size structure. These effects are not nuisances to hide; they are part of the model and must be included in the baseline ladder. The fifth layer is reference-solver uncertainty. The FD-reference sequence shows that passing a circular sanity check is not enough for a shape-sensitive continuum residual. A reference can be useful for calibration and still be too uncertain for the intended residual analysis.

Transport adds one more layer: contacts. Conductance is not only a property of the isolated dot shape. It depends on the coupling between the finite dot and the leads, on lead width, and on mode structure. The preflight therefore does what a preflight should do: it blocks a large shape-transport campaign until the lead-dot coupling baseline is controlled.

These lessons also clarify the role of ML. A surrogate can be valuable for compression, interpolation, and candidate generation, especially when direct Kwant calculations are more expensive than model evaluation. But in the present dataset, a flexible model is not automatically more scientific than a Ridge baseline tied to confinement descriptors. ML should enter after the physical question survives baseline subtraction, not as a substitute for that survival.

The resulting baseline hierarchy is:

1. check size, area, perimeter, and aspect-ratio scaling;
2. check isotropy, symmetry, and degeneracy;
3. check lattice boundary realization, pixelation, and effective radius;
4. check reference-solver uncertainty before using residuals;
5. check contact geometry before interpreting transport;
6. only then use ML or inverse screening for positive claims.

Applied to the existing outputs, this hierarchy supports a reproducible negative benchmark rather than a new device-design rule.

V. CONCLUSION

This draft summarizes a baseline-first limitations audit for finite superelliptic quantum dots on a square tight-binding lattice. The system remains useful as a controlled

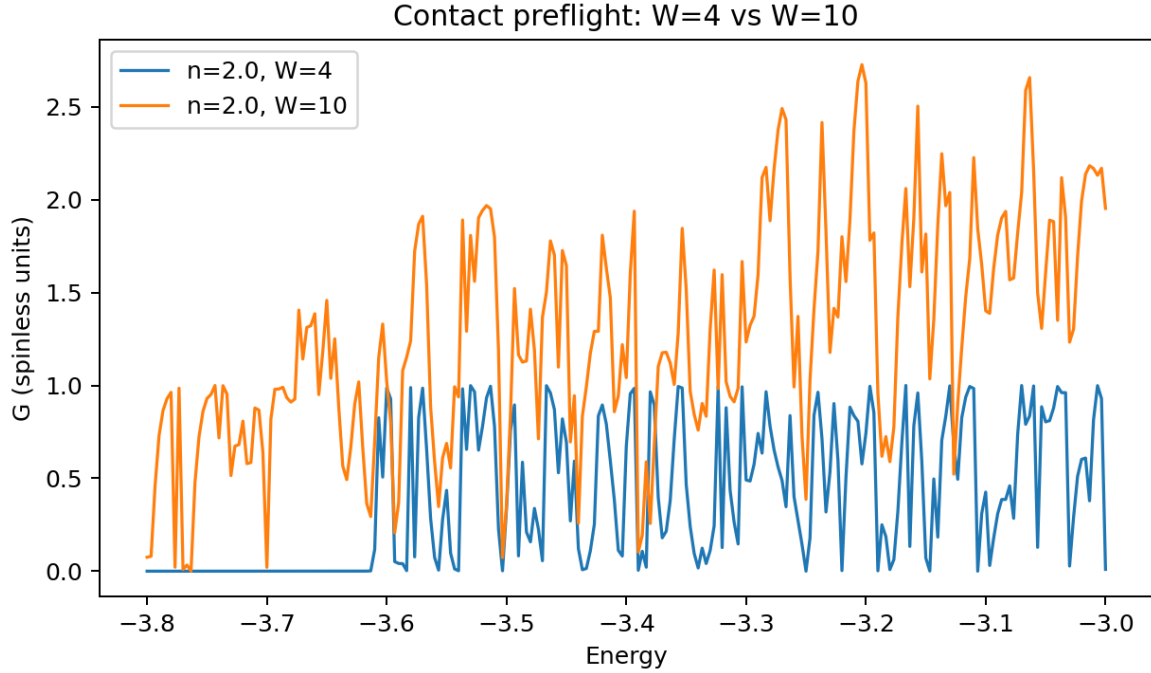


FIG. 2. Existing open-transport preflight plot copied from `reports/article_d_open_transport/figures`. It illustrates the contact-width comparison used to diagnose lead-coupling dominance. The plot is reused as an existing result; no new transport scan was run for this article draft.

TABLE I. Summary of tested hypotheses and limiting baselines. The wording is deliberately conservative: each row reports what the existing result can safely support, not what a stronger future protocol might find.

Direction / hypothesis	Observable	Main baseline or methodological limitation	Verdict	Safe conclusion
Closed-spectrum Ridge/MLP	E_0, dE_1	Physics-informed Ridge and structured validation	ML advantage not established	Ridge remains the preferred conservative surrogate; direct
Q inverse screening	$dE_1/(E_0 + 4)$	Isotropy and same- n baselines	Not supported	Kwant remains the reference. Q candidates do not justify
S objective	$(E_2 - E_1)/(E_0 + 4)$	Simple anisotropy and strongest feasible baselines	Not supported	inverse-design claims. S behaves as an anisotropy
Magnetic ranking crossover	Weak-field level rankings	Symmetry lifting and zero-field/aspect-ratio ranking	Baseline limited	diagnostic in the tested domain. Magnetic response exists, but no
TB-vs-FD continuum residual	Shape-sensitive residual to FD reference	Embedded-mask FD uncertainty	Inconclusive	non-baseline design signal survives. Current FD reference should not
TB-only self-scaling	Ground-state finite-size scaling	Unstable residual exponent and insufficient surviving signal	Inconclusive	be treated as continuum truth.
Open transport contact preflight	$G(E)$ curves	Contact-width dependence	Contact dominated	The fallback gate does not support a positive Path B claim. Transport interpretation requires tighter lead-dot coupling control.

benchmark: the Hamiltonian convention is clear, the geometry family is compact, and the direct Kwant calculations provide a consistent reference for the tested finite systems. The audit shows, however, that several attractive objectives fail to support positive claims once simple baselines and methodological checks are applied.

For the closed spectra, the physics-informed Ridge model remains the preferred conservative surrogate, while a small MLP does not meet the robust structured validation criterion. Q and S inverse-screening objectives do not beat their isotropy, symmetry, or anisotropy baselines. The magnetic ranking-crossover check is explained by symmetry-lifting baselines. The embedded-

mask finite-difference reference is not adequate as a high-precision continuum truth for shape-sensitive residuals, and the TB-only self-scaling fallback remains inconclusive. In the open-transport preflight, contact width dominates the tested shape comparison.

These results do not prove a universal no-go theorem for geometry-based quantum design, machine learning, continuum references, or transport objectives. They show that this particular superellipse metric search does not justify a positive design rule. Future work should therefore change the physical question or the experimental/numerical protocol rather than continue minor variants of the same objective. A stronger future direc-

TABLE II. Selected numerical gate values used in the audit. All entries are taken from the source inventory and existing repository outputs.

Gate	Key value(s)	Internal verdict	Article-level use
Diploma closed-spectrum scale checks	Bessel-scale error about 2.03%; fixed-shape scaling deviation about 2.12%	calibration checks passed for thesis scope	Establishes controlled benchmark, not new shape physics.
Ridge versus small MLP	Ridge relative MAE 2.01–2.60% for E_0 and 0.78–1.15% for dE_1 ; MLP success 10/16 cells	robust MLP criterion not met	ML advantage not established.
FD N501 reference gate	circle raw N501 error 0.00267; non-circular relative uncertainties 0.00472 and 0.00219	FD N501 inconclusive	FD is not used as shape-sensitive truth.
TB self-scaling gate	best $n = 2.0$ Bessel relative error 7.21×10^{-4} ; p_{TB} unstable	TB self-scaling inconclusive	No positive Path B claim.
Open transport preflight	contact effect 0.659; shape effects 0.581 and 0.512	contact dominated	Contact width dominates the tested setup.

tion would need to define the baseline ladder in advance, control reference uncertainty and contact geometry, and

make positive claims only after the surviving signal is direct, stable, and not absorbed by simpler physics.

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